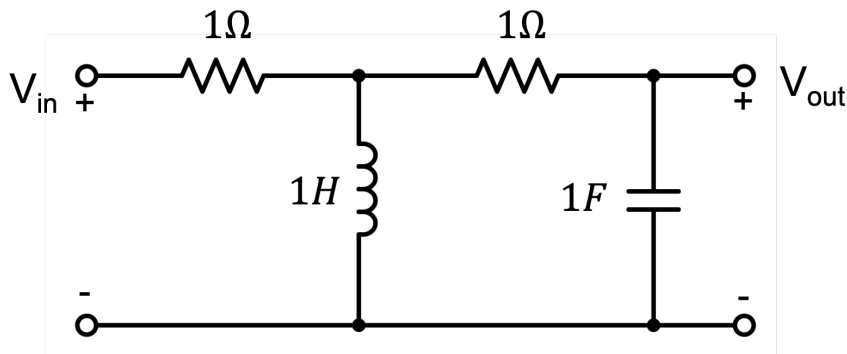


For the circuit below, $V_{in}(t) = 1 \text{ V}$ for $t < 0$, and $V_{in}(t) = (1 + \cos(t)e^{-t})u(t) \text{ V}$ for $t > 0$



- (10 points) Calculate the initial capacitor voltages and inductor currents
- (10 points) Draw the s -domain representation of the circuit.
- (10 points) Find the s -domain representation of the input $V_{in}(t)$ for $t > 0$.
- (20 points) Write mesh current equations in the s -domain that could be used to find the current through the capacitor. Use these equations to fill in the entries in the Cramer's method matrices. Use I_1 and I_2 mesh currents, V_{in} voltage sources.
- (15 points) Find the output $V_{out}(s)$ in the Laplace domain.
- (15 points) Find the output $v_{out}(t)$ in the time domain.

2. (20) Here is the Matlab code for finding the output of an RC circuit to a given input. Change the input to $x(t) = u(t - 2) - u(t - 4)$, and plot the output $y(t)$.

```
RCPulseExample.m * x +
1  % This example needs the Symbolic Math Toolbox of MATLAB
2  clear; % Start by clearing the workspace
3  syms t s X(s) H(s) Y(s) x(t) h(t) y(t); % Declare the symbols
4
5  X(s) = 1/s-exp(-s)/s; % specify the input X(s)
6  H(s) = 2/(s+2); % specify the system H(s)
7  Y(s) = X(s)*H(s); % System output in the Laplace domain
8  x(t) = ilaplace(X); % inverse Laplace transform x(t)
9  y(t) = ilaplace(Y); % inverse Laplace transform y(t)
10
11 % Plot the results
12 time = 0:0.01:5; % array of time points for plotting
13 figure; plot(time,x(time),time,y(time),'LineWidth',2);
14 xlabel('t(sec)','FontSize',12,'FontWeight','bold');
15 ylabel('Voltage (V)','FontSize',12,'FontWeight','bold');
16
17 % Pause for the user and close the figure
18 pause;
19 close all;
```

```
% This example needs the Symbolic Math Toolbox of MATLAB
clear; clc;
% Start by clearing the workspace
syms s t X(s) H(s) Y(s) x(t) y(t)

% Define the input x(t) = u(t-2) - u(t-4)
x(t) = heaviside(t-2) - heaviside(t-4);

% Take Laplace transform of x(t)
X(s) = laplace(x(t), t, s);

% Define the system H(s) for RC circuit (example: 2/(s+2))
H(s) = 2/(s+2);

% Compute output in Laplace domain
Y(s) = X(s)*H(s);

% Inverse Laplace transform to get y(t)
y(t) = ilaplace(Y(s), s, t);

% Plot the result
time = 0:0.01:10; % array of time points for plotting
figure;
fplot(y(t), [0 10], 'LineWidth', 2);
xlabel('Time (s)', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Output y(t)', 'FontSize', 12, 'FontWeight', 'bold');
title('Response of RC Circuit to Pulse Input', 'FontSize', 12, 'FontWeight', 'bold');
grid on;
```

